Adaptive LIVing Engineered organs (ALIVE)

1. General information

1.a Applicants

Name	*Institution, faculty*	*Department*	*Email*	*Tenured / Tenure-track/ Other (specify)*	*For Erasmus MC applicants: clinician/ researcher*	*Role in Flagship*
Amir A. Zadpoor	**TU Delft**, 3mE	Biomechanical Engineering	a.a.zadpoor@tudelft.nl	Professor		**Main Lead**
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Corresponding Lead: Amir A. Zadpoor
Planned start date of seed funding: *01/09/2022*
Primary Convergence theme: *Human Centered Technology and AI for Health*
Secondary Convergence theme (*if applicable*): *Fundamentals of Health & Disease*
Tertiary Convergence theme (*if applicable*): *Technology Supported Transitions in Health(care)*
Quaternary Convergence theme (*if applicable*): Choose an item.
Keywords: *Tissue Engineering; (bio)printing; bio-sensing; transplantation, human disease models*

1.b Summary (max 300 words)

Throughout life, our tissues and organs can fail or stop functioning properly. In most cases we do not have adequate solutions to repair them. Moreover, there is a severe worldwide shortage organs and tissues for transplantation. Despite progress made in the field of tissue engineering and regenerative medicine over the last decades, it remains challenging to generate functional, multi-cellular tissues in the laboratory. To overcome these challenges, we will develop a technological toolbox that will enable the groundbreaking feature of "adaptive spatiotemporal control" in order to generate functional Engineered Living Materials (ELMs). The toolbox will enable spatiotemporal monitoring of cells in ELMs via (bio)sensors and controlling their behavior by delivering the right stimuli to the right location via (micro)channels that will be co-printed into ELMs. Computational algorithms (artificial intelligence) will be used to decide the appropriate response (*e.g.*, delivery of growth factors, mRNA, peptides, oxygen, or nutrients) to control the development process. This way we will be able to mimic multi-staged processes that are required for tissue development in the laboratory. An adaptive infection prevention system will be designed and incorporated to keep the ELM infection-free over time. As proof of concept, we will create three multi-cellular tissues with complex 3D architectures: *(1)* liver, *(2)* osteochondral (bone-cartilage unit), and *(3)* innervated muscle. The final aim is to develop closed-loop automation systems to develop ELMs that can be used for laboratory models to study disease mechanisms,develop treatments and for the replacement of severely damaged tissues. While having a high potential for scientific and technological breakthrough, the application for healthcare faces ethical, legal, economic, and educational issues. We will develop responsible innovations using co-creation and engagement with different stakeholders (*e.g.*, patients, medical doctors, regulatory bodies, companies, students) for different applications of ELM in healthcare.

1.c Public Summary (max 300 words)

Onze weefsels en organen kunnen beschadigd raken en daardoor niet meer goed functioneren. In de meeste gevallen hebben we geen adequate oplossingen om dit te herstellen. Bovendien is er een tekort aan organen en weefsels voor transplantatie. Met "Tissue Engineering" willen we nieuwe weefsels maken in het laboratorium. Ondanks de vooruitgang in de afgelopen decennia, blijft het een uitdaging om functionele, meercellige weefsels te maken. Om deze uitdagingen het hoofd te bieden, gaan we technologie ontwikkelen om het gedrag van de cellen tijdens het maken van nieuw weefsel in het laboratorium beter te controleren en te sturen. Hiervoor zullen we hele gevoelige (bio)sensoren aanbrengen op verschillende plaatsen in het weefsel in hele kleine kanaaltjes die we maken met behulp van 3D-printen. Computeralgoritmen zullen worden gebruikt om te bepalen wat we moeten toedienen via die kanaaltjes om de ontwikkelingsprocessen in de verschillende gebieden in het weefsel te beheersen. Zo kunnen we meertraps-processen die nodig zijn voor weefselontwikkeling, nabootsen in het laboratorium. Er zal een adaptief infectiepreventiesysteem worden ontworpen om het weefsel in de loop van de tijd infectievrij te houden. Als eerste zullen we drie meercellige weefsels creëren met complexe 3D-architecturen: *(1)* lever, *2)* kraakbeen-bot, en *(3)* spier met zenuw. Het doel is om geautomatiseerde systemen te ontwikkelen om weefsels en organen te maken die kunnen worden gebruikt als laboratorium-modellen om ziektemechanismen te bestuderen en behandelingen te ontwikkelen, en voor transplantatie om ernstig beschadigd weefsel te vervangen. Hoewel dit concept veelbelovend is voor wetenschappelijke en technologische doorbraken, zijn er niet alleen technische maar ook belangrijke maatschappelijke uitdagingen, zoals gebruik van weefsels en toegankelijkheid van zorg. Om met onze aanpak verantwoorde innovaties ontwikkelen voor verschillende toepassingen in de gezondheidzorg, zullen we ethische, juridische, economische en educatieve aspecten onderzoeken met verschillende belanghebbenden zoals patiënten, artsen, studenten, regelgevende instanties en bedrijven.

2. Relevance to Convergence Health & Technology mission

(50%) – **max. 2500 words for 2.a to 2.e**

2.a Contribution to H&T mission and fit with H&T theme(s)

This proposal is perfectly aligned with the mission of Convergence Health and Technology and in being so, fits with three of the main themes as outlined below. ALIVE will create and combine new technologies such as 3D and 4D (bio)printing and new sensor technology to provide spatiotemporal control using Artificial Intelligence (AI) that is beyond state-of-the-art to develop Engineered Living Materials (ELMs).

The ELMs developed will be used as clinically relevant models for the study of disease mechanisms. They will also be used in the development and evaluation of new drug-based treatments for prevention and early diagnostics with the aim of maintaining health throughout life. This fits perfectly with **Theme 1: Fundamentals of Health and Disease. Using patient-derived material (cells),** these human models will **enable development of early and more efficient individual-tailored treatments**, preventing more serious disease development and improving **life-long health and wellbeing**. It will also be possible to use ELMs to replace damaged tissues and organs, solving the issues with shortage of tissues and organs for transplantation and providing tissues and organs from a patient's own cells. This will obviously reduce the need to use life-long immunosuppressive medication.

We will use a completely new approach to tissue engineering based on spatiotemporal sending and receiving of signals that steer the development and functioning of living materials *in situ,* while monitoring for and preventing infection of the new tissue. Multi-material 3D and 4D (bio)printing processes, sensors and embedded electronics will be applied to generate these tissues with built-in monitoring and control systems, performed using AI. This approach is what is envisaged in the description of **Theme 2: Human Centered Technology and AI for Health.**

We will focus on tissues of the musculoskeletal system (cartilage, bone and muscle) as well as liver, with the inention of applying **our new technologies to other tissue types** in the future. ALIVE aims for a high impact on the socio-economic burden of the **cost associated with the chronic treatments and potential surgeries patients** have to go through. This effectively has an impact on the quality and length of people's lives. This ambitious and highly innovative project will make Tissue Engineering and Regenerative Medicine (TERM) a pillar of the existing healthcare systems through establishment of an integrated technological platform. This will enable development of personalized regenerative treatments taking into consideration ethical, legal and financial aspects and designing pathways for responsible implementation simultaneous with the development of the technology, thereby addressing **Theme 4: Technology Supported Transitions in Healthcare.**

2.b Synergy between disciplines

The field of TERM is built upon interdisciplinary research between engineers, biologists, clinicians, industry, ethicists, and policy makers. In this programme, all partners will collaborate to develop, and apply innovative and impactful technologies for the generation of new, more complex, tissues and organs.

To enable tissue engineering of complex structures, bioprinting is required to deposit cells in specific areas in a material. We will apply technology that was originally developed for implants, now for cell-based applications: 3D- and 4D-printing (expertise dr. Zadpoor-TUD), stimulus-responsive biomaterials (dr. Apachitei, dr. Fratila-Apachitei–TUD) and design of scaffolds using multi-materials (dr. Zhou, dr. Mirzaali-TUD).

For the generation of liver tissue, these technologies will be combined with the knowledge on different cell types (dr. van der Laan-EMC) and extracellular matrix components (dr. Verstegen-EMC) to 3D-bioprint co-cultures of spatially organised cell types. For the generation of osteochondral tissue, expertise in tissue engineering and pathobiology of cartilage (dr. van Osch-EMC) will be combined with tissue engineering of bone (dr. Farrell- EMC) and knowledge of bone metabolism (dr. van der Eerden-EMC). 3D- and 4D-bioprinting will be used to control the properties of the different tissue layers. To engineer innervated muscle, neurological knowledge (dr. de Zeeuw-EMC) will be combined with the expertise on generation of muscle from stem cells (dr. Pijnappel-EMC) and the 3D-bioprinting technologies.

Generation of complex tissues requires precise spatiotemporal control of the tissue engineering process. To allow sensing and control on-the-fly during the generation of tissues in culture, engineers (group dr. Zadpoor-TUD) will use 3D-printing technology to make a microchannel system from which to draw samples. For sensitive sensing, the technologies developed for single peptide sequencing sensors (dr. Dekker-TUD) will be incorporated. The use of artificial intelligence (AI) (dr. Hunyadi, dr. Tumer-TUD) will be applied for autonomous decision making on the best action based on the results of the sensors.

Since long-term culture in the micro-channel device is required to obtain ELM, we have included microbiological expertise (dr. van Wamel-EMC) to co-develop novel methods for sensitive detection and combat of infections.

Finally, to develop sustainable technologies, we will add new dimensions from ethics (dr. Bunnik-EMC), academic-industry partnerships (dr. Tasselli-EUR), law (dr. Buijsen-EUR), economics (dr. Redekop-EUR), business innovation (dr. van den Ende-EUR) and social sciences (dr. Hulst-EUR) to facilitate and enable implementation of the ELMs to the benefit of patients. Together with the engineers and scientists and through our industry collaborations and connections with clinical colleagues, we have developed a well-rounded team that can and will address the most pressing issues to develop and implement ELMs in line with stakeholders' needs and preferences.

2.c Long term sustainability plan

Feasibility:

- The work proposed here is cutting edge in each of its respective fields, but the real novelty lies in the combination of this expertise in a truly interdisciplinary manner.
- This project builds and expands upon existing collaborations between several research groups within TU Delft and Erasmus MC. Some of these collaborations exist over 10 years, such as on bone and implants (Farrell and van der Eerden with Apachitei and Fratila-Apachitei) and the collaborations on bioprinting between van Osch, van der Laan and Zadpoor were initiated 3 years ago. In recent years, collaborative work has led to many joint publications as well as applications for national and international funding. This forms a strong basis of a solid and sustainable collaboration.
-Expansion of this core group brings in further expertise in a range of research fields, such as innovative sensor technologies, other tissues upon which to apply our technology (*e.g.*, innervated muscle) and the incorporation of insights from ethics, law, economics and business innovation throughout the development process. These add to the core skillsets already well established and bring all necessary expertise to the project.
-The vast experience with multidisciplinary collaborations of most of the PIs but in particular also the three project leads, will facilitate the success of the project.

Sustainability:

- To be sustainable beyond the life of this flagship, it is crucial to carry out work that is timely, relevant and novel. It is clear from the research outputs and recognition in the respective fields that

this team is carrying out extremely relevant research. Combining this expertise will consolidate that position.
-The topic of ELMs is timelier than ever, given the clear need for the development of new approaches for the repair and replacement of most if not all tissues and organs. In this programme, we aim to jointly develop entirely new technologies that will enable overcoming some key barriers in the field of TERM. The ambitious nature of the work proposed here ensures not only that this work will still be relevant in 5 years but also that it can easily be expanded and applied to many other tissues and organ systems.
- A sustained financial plan to guarantee the programme after the initial 5 years will be established by continuously accessing national and international funds relevant for the specific themes addressed in the project (*e.g.* Zwaartekracht, NWA-ORC, TTW-Perspectief and Horizon Europe). Matching is already available through relevant ongoing projects. More co-funding from NWO (TTW in particular), different HORIZON-Europe programmes as well as via public-private collaborations will be applied for in the coming 5-years. Partners involved in this programme have long lasting track records on the topic with external funding from NWO, ERC, TTW, KNAW, Health-Holland/TKI, Interreg, HORIZON (*e.g.*, EIC, MSCA, FET, societal challenges) as well as funding from Dutch Arthritis Association, AO Foundation, KWF Dutch Cancer Society, Dutch foundation for Liver and Gastroenterology and Princes Beatrix Foundation.
-To strengthen the connections and create the environment for long-term collaborations, we will set up a virtual Center for Bioinspired Synthesis, Assembly, and Fabrication. This is initiated from an existing regional collaboration between the groups of Zadpoor, van Osch and van der Laan in the Medical Delta. This initiative would also be connected with the currently ongoing plans for a Biofabrication facility.

Main risks for viability and sustainability, and mitigation strategy:

Risk	**Mitigation strategy**
Technical difficulties leading to blockage in progress- *medium risk*	Promising pilot data present; back-up plans for new technology available
Insufficient funding to match and/or continue the programme- *low risk*	Programme has established PIs with proven track record to attract external funding as well as high potential young PIs
Disagreement between partners on decisions related to strategy- *low risk*	All partners involved have experience with multidisciplinary collaboration in consortia. The governance structure will help to solve issues and to prevent a collapse.
Withdrawal or loss of key PIs from the programme – *low risk*	A large group of PIs ranging from full professors to tenure trackers is available. Overlapping expertise is present in and between the PIs.

2.d Visibility, attracting talents and new partners

<u>Visibility:</u>
- We will create a website and a logo for the Flagship that will be linked to the websites of the PIs involved.
- In oral and poster presentations at local and (inter)national meetings, the PIs as well as the personnel appointed to the project will actively promote the Flagship programme.
- Our strong involvement in the international community will help further advertise the successes of this consortium.

Engaging students and early career researchers:
- Multidisciplinary collaboration requires a different attitude. Making students aware of differences in ways of approaching a problem as well as different societal, medical and technical challenges and opportunities will lead to the creation of the multidisciplinary researchers of the future. PIs of ALIVE have introduced multidisciplinary education in TERM in Rotterdam and Delft and over the last 3 years have expanded this to cover bachelors, masters and PhD courses. This will create a new generation of students that can be actively involved in the ALIVE programme.
-The research groups within this project have been proven to be extremely active in attracting Masters and Bachelors students for their internship and research projects and regularly host students from the counterpart universities in their labs. Indeed, several joint publications involving Masters students have resulted from this process.
- There is ample experience in training students in interdisciplinary projects. Prof. van Osch has coordinated two Horizon2020 innovative training networks and she, as well as dr. Farrell, dr. van der Eerden have participated in several of these networks.
- We will actively promote the interdisciplinary work of this flagship and this team through our usual channels to continue to attract the best students to our team. The PIs involved in this flagship proposal are highly visible within several Bachelors, Masters and graduate student programmes (see section 4a) in Erasmus MC, EUR and TU Delft where we will specifically stimulate multidisciplinarity and highlight ALIVE as example.

Attracting talents, partners and funding:
-The (inter)national recognition and the networks of our research groups facilitate the attraction of talented students and postdocs as well as potential partners, both from other universities and research institutes and from industry. Through the increased visibility of this flagship as well as the increased critical mass we have generated as a group, we will further attract new talents in the form of students and postdocs.
-Most of the applicants have been actively participating in public-private partnership programmes (*e.g.*, FES, TIPharma, TTW Perspectief, TKI/Health Holland and several European programmes). With the increased visibility of this flagship, we will further attract companies' interest in our work. We have already engaged with industrial parties and start-ups (section 4.c) with 5 letters of commitment and over €100,000 in kind contributions from companies obtained. We envisage that through our new technological developments, we will be of increasing interest to industrial partners and international collaborators in the academic sector.

2.e Long term societal impact

The availability of functional multi-cellular/multi-tissue structures created here offers societal impact through several achievements. We will:
1) Provide the next generation of human disease models to address unmet clinical needs. We will first focus on the treatment of degenerative diseases affecting cartilage and bone (affecting over 40 million people in Europe[1]), innervated muscle tissue, such as in amyotrophic lateral sclerosis (ALS; worldwide prevalence 4.42 per 100.000 population with average survival after diagnosis of 3 year[2]), Pompe disease (PD: incidence infantile onset approx. 1:52.000 -1:140.000 live births; incidence late onset 1:60.000[3]), and chronic and acute liver diseases (*e.g*, cirrhosis, chronic hepatitis, fatty liver disease, and hepatocellular carcinoma; affecting over 29 million patients in Europe[4]). Our new models will be compared with the state-of-the-art models that are currently available in the laboratories of the PIs. We will promote these new models, making them available for other groups via presentations at international conferences, open access publications, hosting scientists from other groups to transfer the knowledge as well as by actively approaching pharmaceutical companies.

2) Address the lack of organ donors (over 150,000 patients in Europe are on an organ waiting list, part of them for liver, and annually only 34.000 receive a transplant (WHO-ONT Global Observation on Donation and Transplantation)). To facilitate the use of ELMs in clinics, the ethical, legal and economic barriers[5] will be minimized by involving stakeholders since the beginning of their development.
3) Offer potential future treatment solutions for patients that currently have no options, such as those suffering from amyotrophic lateral sclerosis (ALS) with only one registered treatment available (Rilutek) that has marginal effects and unclear working mechanism. Moreover, knowledge obtained from ALS studies is applicable for other neurodegenerative diseases, in particular forms of dementia.
4) Reduce the number of laboratory animals (>300,000 in Europe alone) that are killed every year. For example, for hepatotoxicity research, animal studies are the only option now. Functional multi-cellular liver tissue will change this while also improving the relevance of the obtained results to humans.
5) Educate the next generation of scientists, engineers, policy makers, regulators and medical doctors by expanding already established educational programmes (minor, master, PhD) in TERM (at TUD and EMC), including TERM in the master Medical Business and Innovation (at EUR) as well as active lecturing in (inter)national courses to advocate convergence principles.
6) Generate a considerably sized cluster of high-level research that will improve our educational programmes, and that will attract national and international students and scientists.
7) Create an incubator for new technologies. Our cutting-edge technologies will be attractive for (start-up) companies in the region.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

The ALIVE programme aims to implement beyond the state-of-the-art technology to stimulate and improve the **production of tissues and organs with complex 3D architectures and predictable functioning.** This will address medical needs related to the shortage of human tissue for transplantation as well as the availability of reliable human models to develop treatments for diseases that currently have no cure.
Despite the huge potential of TERM to address this shortage of tissue, it remains technically challenging to generate tissues with complex 3D architectures and physiology. Progress has been hampered by lack of robust design principles and technological platforms that allow tight monitoring and control of the tissue development process, in particular when multiple cell or tissue types are to be combined. In ALIVE, we will introduce, for the first time, a combination of (bio)printing technologies with **adaptive spatiotemporal control technology** to overcome these limitations (from Technology Readiness Level (TRL) 1/2 to TRL3/4). The primary novelty of the **ELMs** developed here lies in **automated**, **active monitoring and control** mechanisms that guide both the transient (*i.e.,* initial stages of the development where many adjustments need to be performed) and steady-state (*i.e.,* later-stage, less adjustments are required) phases of developing living materials. Thus, we apply on-the-fly responses to the uncertainties that are inherent in the generation and functioning of living materials, including patient-specific variability.

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3.b Aims and feasibility

We have formulated the following specific research **aims and outputs for the first 5 years**:

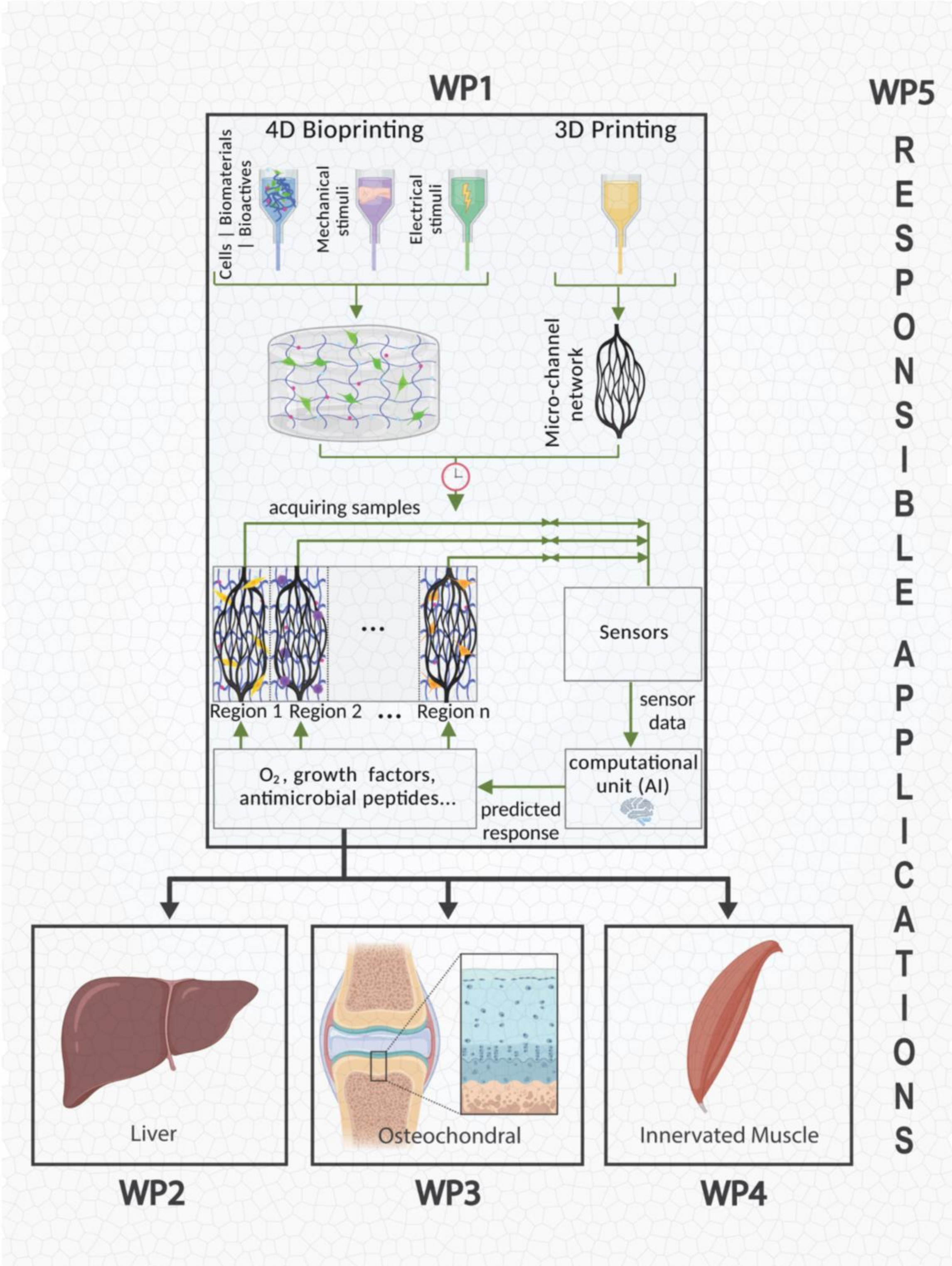
1. A toolbox with key breakthrough technologies that can be used for generation of tissues and organs *(WP1; Postdocs 1-4; 7 person-years)*. The toolbox will be versatile and used for adaptive spatiotemporal control of different engineered tissues/organs. It will consist of: *Bioprinting* techniques to design tissue architecture with properties that can be adapted in time with external physical stimuli (i.e. 4D-bioprinting); *co-printed spatially resolved networks/channels within the tissues* for sending and receiving signals to monitor and control cell behavior and tissue formation with extreme sensitivity and precision; An integrated *infection-prevention* system to sense and prevent infections during tissue development; Predictive models based on *AI* to be used as an "automated brain" directing the decision-making process of control based on sensory information from (semi)real-time sampling.

2. Methods to create complex, functional tissues (*WP2-4; PhD students 1-3; 12 person years & Postdocs 1,2,4; 2 person-years).* We will use the tools generated in WP1 to create three different tissue types with specific complexities regarding their physiology: 1) *liver* that consists of different cell types that have complex paracrine interaction and roles in disease development (WP2); 2) *osteochondral tissue* that consists of three layers of tissue with very distinct structural and mechanical properties (WP3); 3) *muscle* tissue that is functionally *innervated* evidenced by its capacity to sense and generate a signal (*i.e.,* coordinated contraction) (WP4).

3. Pathway for responsible development and implementation of breakthrough technologies *(WP 5; Postdocs 5,6; 3 person-years).* Throughout the entire programme, we will consider *health economic, ethical, and legal issues,* key to the safe and ethically responsible implementation of our methodologies. We will use co-design and public engagement methods to involve stakeholders from different research backgrounds and from society, including (future) medical specialists, patients, regulators and industry to design and adapt routes for implementation of the technology.

Staff/budget
Postdocs will be appointed in WP1 and 5. These postdocs will work closely together with the PhD students appointed on the biomedical applications (WP2-4). Personnel will work closely together with preexisting staff and students involved in related research and technologies.
For the first two years more than sufficient matching is available (annex 2). For the subsequent years, we will submit joint proposals on this topic. We have already submitted an EIC Pathfinder proposal (€7 million; HorizonEurope) and plan for suitable other calls in HorizonEurope, TTW (Perspectief/OTP) and disease specific funds.

Risk	**Mitigation strategy**
Tools from WP1 not available in time–*low risk*	Several prototypes and preliminary data already available. Use of existing technology possible for initial experiments.
Generating complex tissues more difficult than anticipated–*medium risk*	Vast experience among PIs. Tissues will be broken-down into component parts and then assembled using new technologies.
Roadmap to commercialisation unclear-*low risk*	Involvement of experts in business development and regulatory issues combined with industrial input mitigates this risk.


WP1
4D Bioprinting
3D Printing
Cells | Biomaterials | Bioactives
Mechanical stimuli
Electrical stimuli
Micro-channel network
acquiring samples
Sensors
sensor data
Region 1
Region 2
...
Region n
O_2, growth factors, antimicrobial peptides...
computational unit (AI)
predicted response
Liver
Osteochondral
Innervated Muscle
WP2
WP3
WP4
WP5
RESPONSIBLE APPLICATIONS

3.c Work plan

WP1. **Development of a technological toolbox for adaptive spatiotemporal control (WP leader: Fratila-Apachitei;** co-applicants Mirzaali, Zhou, Tumer, Dekker, Hunyadi, van Wamel, Apachitei, Zadpoor**)**

Task1.1. 3D / 4D (bio)printing process: We will produce 4D-printed[6] cell-laden (hydrogel) structures from stimuli-responsive biomaterials to be used to generate ELMs. We will use multi-modal dynamic (2D to 3D) shape-shifting changes over time using origami design concept[7–9] to aid in development of 3D tissues. We will develop a hybrid 3D-printing platform enabling the combined printing of multiple material types in one construct to improve the level of tissue complexity.

Task1.2. Micro-channel device with embedded electronics: We will design and develop high-resolution (micro/nano-)channel networks using Direct Laser Writing or Electron Beam Lithography. These channel networks will be used for high sensitivity *in situ* monitoring (*e.g.* oxygen, RNA, proteins) and controlling tissue formation.

Task1.3. Generation of sensitive sensors: We will capitalise on a recent breakthrough in the Dekker lab[10] to develop fast and automated detection of proteins with extremely low abundance in our tissues with sensors that can sequentially read the amino acid sequence of native proteins.

Task 1.4. Development of an infection prevention system. We will design and implement a closed-loop adaptive infection prevention system: *1)* using materials in the devices with low biofilm formation; *2)* implementing sensors for sensitive detection of microbes based on peptide sequences; *3)* developing methods to combat infections in the ELMs using antimicrobial peptides, lactoferrin immunoglobulins, or bacteriophages.

Task1.5. Automated brain. AI-based approaches will be used for *1)* prediction of cells' responses to various physical and chemical stimuli; *2)* optimal real-time monitoring (*e.g.,* denoising signals coming out of sensors) and spatiotemporal control of ELMs with appropriate stimuli (*e.g.* which stimuli to be sent where and when); 3) determination of infections in the ELMs and prediction of the proper dosage of antimicrobial agents.

WP2. Organ Printing: Multi-cellular tissues with complex cellular interactions (WP Leader Verstegen; co-applicants van der Laan, Dekker, Fratila-Apachitei**)**

Task2.1. Select bioinks for the different cell types to generate liver tissue: Use available analyses of the liver's extracellular matrix composition to select and fine-tune hydrogel-based bioinks[11,12] that support the design, mechanical cues and growth factor requirements of liver epithelial cells (cholangiocytes, hepatocytes), endothelial cells, fibroblasts and Kupffer cells.

Task2.2. Bioprint spatially organised tissue and incorporate physical/chemical cues: To mimic the liver architecture, we will create specific printing patterns. Typical zonation/gradients of nutrients and oxygen found in liver lobules that provide differentiation cues will be incorporated using the micro-channel system.

Task2.3. Mimic diseases and develop biomarkers to allow spatiotemporal control: Via the printed micro-channels, we can induce disease processes by targeting a specific cell type (*e.g.,* high concentration lipids to mimic fatty liver disease or viral infections for hepatitis E). The response of the targeted cell type and the response of the other cells will be evaluated separately, with sensors in the specific microchannels.

WP3. Generating mechanically functional interfaces: osteochondral tissue construct (WP leader Farrell; co-applicants van der Eerden, Mirzaali, Tumer, van Osch**)**

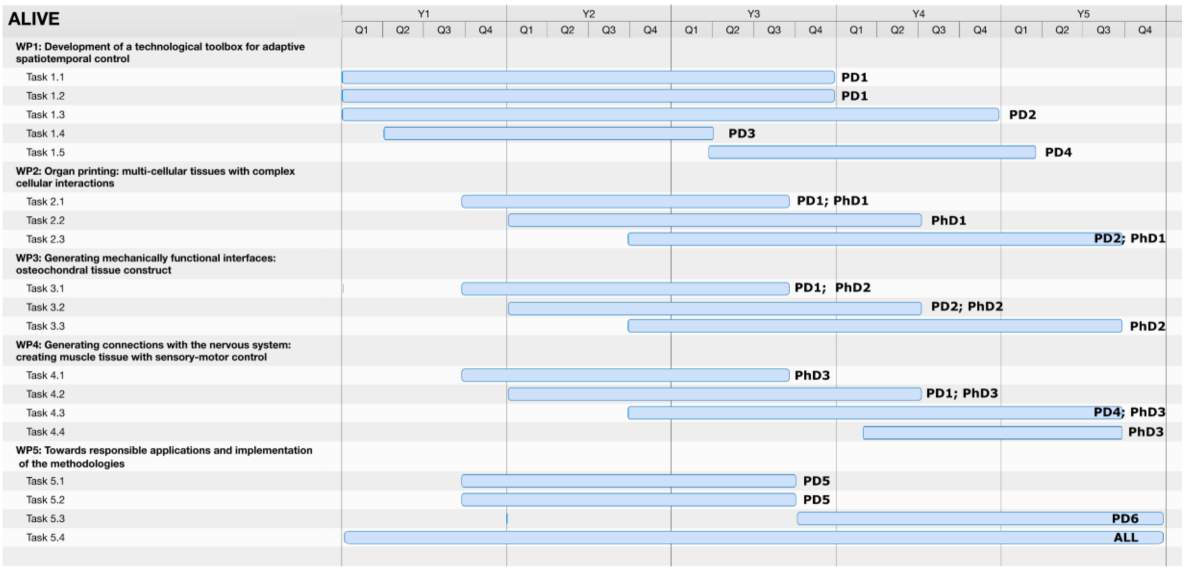
Task 3.1 Generate the interface layer: To date we can tissue engineer bone via the process of endochondral ossification using mesenchymal stem/stromal cells (MSCs) and stable cartilage using chondrocytes[13,14]. A main challenge in the formation of the complete osteochondral unit is creating the interface layer: the calcified cartilage. We will focus on nutrients, oxygen, and growth and

mineralisation factors (*e.g.,* pyrophosphate, ANK, IHH, ENPP1, MGP, fetuin, TGFb, FGF, SPARC, nano-hydroxyapatite) to control cell fate in this layer. We will assess rate of mineralisation, matrix production and cellular phenotype using either MSCs, chondrocytes or a mixture.
Task 3.2. Identify markers and incorporate cues to direct/control cell behaviour: We will identify markers for different cell fates and tissue maturation (*e.g.,* calcium, ALP, IHH, COLX). The three layers will be monitored and controlled in a spatiotemporal manner through sensing and delivery via the microchannels.
Task 3.3 Generation of the osteochondral unit *in vitro* and testing *in vivo*: We will bioprint the 3 layers of the osteochondral unit (>1cm^3) with microchannels. Mechanical (nanoindentation; bulk) and histological (glycosaminoglycans; collagen types) properties will be evaluated over time *in vitro*. As ultimate test of viability and stability, the generated ELM will be implanted into a semi-orthotopic osteochondral defect model in mice[15,16].

WP4. Generating connections with the nervous system: creating muscle tissue with sensory-motor control (WP Leader de Zeeuw; co-applicants Pijnappel, Jaarsma, Mirzaali, Hunyadi**)**
Task 4.1. Generation of motoneurons and muscle cells: Different lineages of primary murine and iPSC-derived human motoneurons and skeletal muscle cells with different sensitivities for neuromuscular degeneration (*e.g.*, ALS, PD) will be characterised.
Task 4.2. Spatiotemporal control of cell differentiation and connectivity: To connect muscle and neurons, we will generate a three-part system (cortex-spinal-muscle combinations[17]). For this, we will create a variety of assemblies using the different motoneurons and contractile 3D tissue engineered skeletal muscles[18], and use optogenetics to activate genes with light to control cell behavior.
Task 4.3. AI to integrate multiple actuation and sensing signals: Function of neuromuscular junctions will be assessed by various readouts (biochemistry, histology, electrophysiology, contractile force[19,20]), revealing complex multivariate datasets. Signal processing and AI will be used to denoise and classify these data (*e.g.,* healthy vs pathological).
Task 4.4. Innervated muscles to study functional and pathological implications *in vitro*: The assemblies will be interrogated with different sensory stimuli to create different patterns of muscle contraction. We aim for different assemblies that will have different sensitivities for neuromuscular degeneration, and that can be used to model ALS and PD.

WP5. Towards responsible applications and implementation of the methodologies (WP leader: van den Ende; co-applicants Bunnik, Redekop, Hulst, Buijsen, Tasselli**)**
Task 5.1. Ethical issues related to the development and use of ELM: We will conduct a literature review on public perceptions of human ELM that can 'sense and act', and employ normative analysis to determine the conditions under which the use of ELM in research and clinical settings is socially and ethically acceptable, *e.g.* in relation to informed consent, donor control, feedback of research results and incidental findings, reduction of animal experiments, tissue governance, (equal) access and affordability, commercialisation.
Task 5.2. Regulatory issues and clinical translation: In collaboration with several stakeholders, including industry, we will identify opportunities and pitfalls in the design of ELM. Legal issues considered are safety of medical treatment with ELM, how to classify ELM, the need to develop another legal regime for 'living medical devices', including requirements and conditions and liability in case of physical and mental injury for the patient.
Task 5.3. Academia-Industry collaborations: We aim at disentangling *1)* patterns of knowledge transfer, *2)* inter-personal and organizational networks, and *3)* relationships between professionals with the overarching goal of answering these questions[21,22]: (*i*) How do labs/science-based companies develop innovation (at the idea generation phase)?, (*ii*) What is the relationship

between different professional groups in this process (*e.g.*, managers vs scientists)?, (*iii*) What is the role of formal organizational structures (*e.g.,* teamwork or flat organization) and informal networks?
<u>Task 5.4. Exploring possibilities and barriers for future use of ELM</u>: Using brainstorm and round table meetings, we will jointly design, monitor and execute a path for the future. This task will be organised by the project manager together with dr. van Osch. The pathway will include the definition of new tissues to apply the techniques on, finding new partners, attracting new funding.



3.d Alignment of the program with relevant national and international research agendas

ALIVE aligns with
- local and regional "Medical Delta" initiative on development of sustainable and technological solutions for health.
- several routes in Dutch National Science Agenda: Regenerative Medicine; Materials-Made in Holland; Personalised Medicine; Measuring and detecting anything, anytime, anywhere.
- NWO programme "more knowledge with fewer Animals".
- when we are about the enter the preparations for clinical application, we will contact RegMedXB (supported by Groeifonds) to discuss the use of their infrastructure.
- European research agenda, the draft of the Health Work Programme 23-24 announced HORIZON-HLTH-2024-TOOL- 5.02: Bio-printing of living cells for regenerative medicine.

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

<u>LEADS:</u>
Prof.dr.Ir. Amir Zadpoor (TUD-3mE)
- Head of Biomaterials & Tissue Biomechanics group.
- Expert in bio-inspired porous implants using (multi-material) additive manufacturing.
- H-index: 68; citations: 14,900.
- Won several personal grants (VENI,VIDI) and led European projects (e.g. ERC-starting grant, Interreg2Seas).
- Active in several committees and editorial boards of scientific journals.

- Director of Biomedical Engineering Education and instructor in several BSc and MSc courses (*e.g.*, Regenerative Medicine, 3D printing, Computational Mechanics of Tissues and Cells).

Prof.dr. Gerjo van Osch (EMC)

- Leading basic/translational research group focussing on cellular processes in cartilage regeneration.
- Expert in tissue engineering and regenerative medicine.
- Co-authored >250 publications; H-index: 57; citations: 10,628.
- Coordinated several multidisciplinary consortia with public-private partnerships including 2 Horizon2020 MSCA-ITN.
- Member of the board of research master molecular medicine; previously coordinator master research for medical students.
- Past-chair of European Chapter of Tissue Engineering and Regenerative Medicine International Society (TERMIS), vice-chair of Gordon Conference Cartilage Biology and Pathology, active in several (inter)national societies, committees and editorial boards.

Prof.dr. Jan van den Ende (EUR-RSM)

- Research interests include firm-internal and -external idea management, role of project leaders and ecosystem innovation.
- H-index: 28; citations: 3,749
- Recently published a textbook Innovation Management .
- Strong industry relations, including as founder of the New Business Roundtable, a network of large corporates such as ING, Philips, Royal Cosun, ASM International.
- Extensive teaching experience, particularly for executive audiences.

<u>APPLICANTS:</u>

Prof.dr. Cees Dekker (TUD-TNW)

- KNAW Royal Academy Professor.
- Trained as physicist and pioneered nanotechnology in the 1990s.
- Now focuses on single-molecule biophysics and nanobiology, including nanopores for sequencing and mimicking nuclear transport, chromosome biology, and synthetic cells.
- Published >370 papers; H-index: 112; citations: 77,187
- Awarded many prizes including 3 ERC Advanced Grants.

Dr. Jie Zhou (TUD-3mE)

- Interested in advanced biomaterials for bone tissue engineering using 3D printing technologies.
- Led or involved in many national and European projects in the past 30 years.
- H-index: 46; citations: 6,145
- Currently managing the Interreg 2 Seas project (3DMed) with 17 European partners (budget of nearly €7million).

Dr. Lidy Fratila-Apachitei (TUD-3mE)

- Focused on development of cell-instructive topographies for generation of tissues.
- H-index: 27; citations: 2,426
- Instructor in several BSc and MSc courses (*e.g.*, Regenerative Medicine, Biomaterials and Tissue Biomechanics).

Dr. Iulian Apachitei (TUD-3mE)

- Interested in the surface biofunctionalization of biomaterials to control material-cell interactions.
- H-index: 28; citations: 2,749

- Coordinator of the Biomaterials and Tissue Biomechanics Specialization in 2016-2017 and the Biomedical Engineering Master Programme in 2017-2021. Instructor in several BSc and MSc courses (e.g., Biomaterials, Study of Materials).

Dr. Mohammad J. Mirzaali (TUD-3mE)

- Assistant Professor, main research interests biomechanics, biomaterials and biomimicry.
- Received several national and international awards and grants including Idea Generator (NWA-IDG) and Open Mind Convergence.
- H-index: 18; citations: 1,125
- Instructor in MSc Courses in Biomedical Engineering programme and supervising students for graduation projects.

Dr.ir. Nazli Tumer (TUD-3mE)

- Interest in medical image processing, development of computational models and mechanical testing.
- H-index: 15; citations: 817
- Instructor in several BSc and MSc courses (Clinical Technology, Technical Medicine, Biomechanical Engineering students) and supervising MSc students, interns and a post-doc.

Dr. Borbála Hunyadi (TUD-CAS)

- Uses biomedical signal processing (EEG, ECG, fMRI, (f)US) and machine learning to tackle pattern recognition problems.
- H-index: 18; citations: 919
- Involved in multiple projects including ME Synergy grant, Medical Delta Cardiac Arrhythmia Lab, and prostate cancer detection TKI-PCaVision.
- Lecturer in Signal Processing and Applied Convex Optimization.

Prof.dr. Chris de Zeeuw (EMC)

- Chair of Dept of Neuroscience
- Received the PIONIER Award of NWO, an ERC-Adv grant and many others (e.g. NWO, ZonMw, FES, HFSP, EU, KNAW).
- Member of the Royal Dutch Academy of Arts & Sciences.
- Published 3 books and >500 papers in neuroscience; H-index: 94; citations: 28,171.
- Many of the technologies Chris developed for academic neuroscience research are commercially available via Neurasmus BV.

Prof.dr. Luc van der Laan (EMC)

- Focused on regenerative medicine applications of organoids for liver diseases and improvement of liver transplantation.
- >200 publications; H-index: 53; citations: 20,549
- Has set up a minor Regenerative Medicine and a PhD course Regenerative Medicine between Erasmus MC and TU Delft with involvement of several of the applicants of this programme

Dr. Eric Farrell (EMC)

- Interested in endochondral ossification, osteoimmunology and *in vitro* and *in vivo* models of bone formation in the context of regenerative medicine.
- Secretary of TERMIS.
- H-index: 29; citations: 3,269
- PI/Coordinator of several competitive EU projects, incl. FET-Open and Marie Curie ITN.
- Guest lecturer at TU Delft since 2007.

Dr. Monique Verstegen (EMC)

- Involved in developing and utilizing organoid technology and extracellular matrix in liver regeneration, liver cancer and tissue engineering.

- H-index: 25; citations: 3,391
- PI on several collaborative projects, funded by ZonMW, ENW, MLDS, KWF.
- Involved in initiating and coordinating courses for PhD Research schools, Minor education, and teaching Medical, Clinical Technology, NanoBiology and Biomechanical Engineering students.

Dr. Bram van der Eerden (EMC)

- Expert on bone generation, mineral metabolism and rare bone diseases.
- H-index: 33; citations: 4,966
- PI on several collaborative projects, including Health~Holland.
- International board and committee positions within ECTS, BMAS, ASBMR, IFMRS.
- Coordinates courses for the Clinical Technology Bachelor and Erasmus MC Graduate School as well as lecturing of Medicine, Clinical Technology and Biomechanical Engineering.

Dr. Dick Jaarsma (EMC)

- Focuses on the generation and in-depth characterisation of mouse models of neurodegenerative disease.
- H-index: 42; citations: 6,028
- Current grants: NWO-BBOL; ZON-MW MEMORABEL; NWO-TTW.
- Teaches (bachelor/masters): Gross anatomy, neuroanatomy, embryology, neurodegenerative disorders.

Dr. Willem van Wamel (EMC)

- Primarily focused on developing new strategies to fight difficult to treat bacterial infections.
- H-index: 41; citations: 5,252
- Current grants: IMI, NCOH.
- Teaching: Clinical Technology, core teacher research master Infection and Immunity.

Dr. Pim Pijnappel (EMC)

- Interested in development of methods for skeletal muscle differentiation from human iPSC.
- H-index: 31; citations: 4,554
- Established 3D skeletal muscle-on-chip technology
- Board member Center for Lysosomal and Metabolic Diseases, United for Metabolic Diseases, European Study Group for Lysosomal Disorders, (co)chair of the skeletal muscle theme group of human Organ and Disease Modeling Technologies.

Dr. Eline Bunnik (EMC)

- Leads PhD-projects on ethics of regenerative medicine/cell-based personalised medicine
- Projects include BRAINMODEL consortium (ZonMw-PSIDER) and VANGUARD consortium (EU-Horizon2020).
- H-index: 14; citations: 626
- Member ethics working group of European Society for Organ Transplantation and works with Rathenau Institute and NEMO Kennislink on societal discussions on the animal as a source of donor organs.
- Teaches students in Medicine, Nanobiology and Phylosophy and healtcare professionals.

Prof.dr. Ken Redekop (EUR-ESHPM)

- >25 year experience in health technology assessment, health services research, observational research and clinical trials
- H-index: 49; citations: 9,544
- Co-authored >200 articles.

- Currently involved in four EU-funded projects that develop and translate innovative ideas about prognostic and predictive biomarkers and AI into viable healthcare products/services.
- Editor-in-Chief of Health Policy and Technology.
- Taught various courses relating to health technology assessment, supervised PhD, master and bachelor students

Dr. Stefano Tasselli (EUR-RSM)

- Research focuses on the interplay between psychological and relational antecedents of organisational functioning.
- H-index: 11; citations: 1,381
- Awarded as one of the '40 World's Best Business Professors under 40' by Poets and Quants.
- Member of the Editorial Boards of Academy of Management Journal and Journal of Management Studies.
- Academic Director of the Master in Innovation Management and of Master in Medical Business and Innovation that will start in August 2022.

Prof. mr. dr. Martin Buijsen (EUR-law)

- Expert in access to healtcare, healthcare reimbursement and care contracts, patient rights, privacy in healthcare.
- 360 Articles (52 academic, 223 professional, 85 for general public) and 29 books.
- Teaches several courses in healthcare ethics, healthcare and law.

Mr. dr Ernst Hulst (EUR-ESHPM)

- Expert in patient's rights, medical risks and liability and disciplinary law.
- 87 Articles (31 academic, 56 professional) and 5 books
- Serves in the Supervisory Boards of care institutes and foundations
- Member of two Medical research Ethical Committees (MEC-U and METC Brabant)
- Teaches several courses on healthcare management, law management and academic skills for students and physicians.

<u>Advisors:</u>

Prof.dr. Joost Gribnau: Developmental biologist.
Dr. Samantha Copeland: Ethics and responsible innovation of emerging medical technologies.
Dr. Koen Bos: Orthopaedic surgeon, treats cartilage/bone diseases.
Prof.dr. Ans van der Ploeg: Pediatrician, treats rare muscle disorders.
Dr. Nadine van der Beek: Neurologist, treats neuromuscular diseases.
Dr. Jeroen de Jonge: Hepato-Pancreatico-Biliary and transplantation surgeon.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

We realise the unique opportunity to carry TERM to the next level by developing a technological toolbox that enables the generation of tissues with adaptive, controllable and predictable properties and functionalities. This is a completely new, multi-faceted research line that would not be possible without the recently developed high-tech 3D/ 4D (bio)printing and AI technologies (Zadpoor's lab-TUD), sensory systems (Dekker-TUD) and machine learning (Hunyadi-TUD) with the in depth biological knowledge of tissue engineering and pathology of cartilage (van Osch-EMC), bone (Farrell, van der Eerden-EMC), liver (Verstegen, van der Laan-EMC), muscle (Pijnappel-EMC) and neural connections (de Zeeuw-EMC) and expertise on infection (van Wamel-EMC).
To allow responsible development and implementation of the innovative technology, experts in ethical (Bunnik- EMC), legal (Buijsen, Hulst- EUR), Health Technology Assessment (Redekop-EUR)

and innovation and business development (van den Ende and Tasselli-EUR) will investigate challenges and opportunities. In addition, advisors provide input on different but related perspectives such as Developmental Biology (Gribnau), philosophy of changes related to technological innovations (Copeland) and clinical perspectives (Bos, van der Ploeg, de Jonge, van der Beek).

4.c Involvement of partners

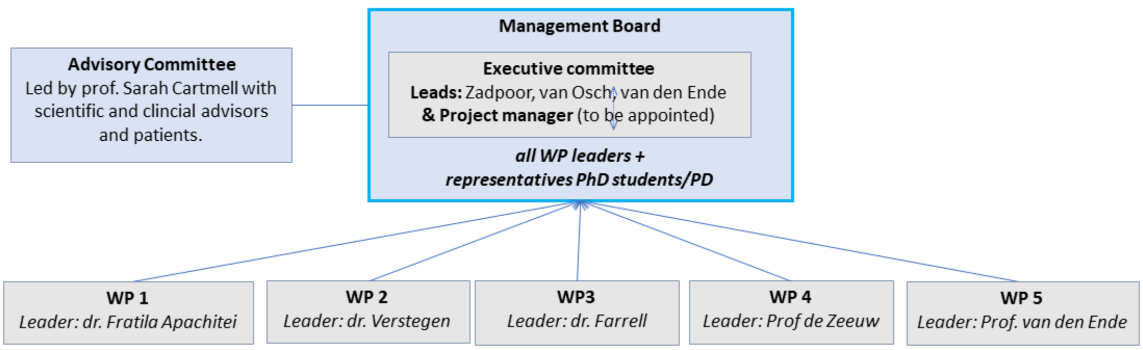
Several companies have expressed interest in this programme. They will share and transfer unique expertise and contribute to educational activities (incl. summer schools and workshops). Combined in-kind contribution: €130,000

- REGENHU: Configure/supply bioprinting instrument and software for liver and muscle bioprinting (task 1.1 and in tight connection with WP2-4).
- Poietis: Develop laser-assisted bioprinting technology for cell deposition at high-resolution up to the single cell ensuring high cell viability (>95%). This technology will be used for the engineering of the multicellular osteochondral interfaces (WP3) and muscle tissues (WP4).
- Nanoscribe/Cellink: Develop advanced bioprinting technologies including extrusion-based and light-assisted techniques at extremely high resolutions which will be used for microchannel fabrication (task1.2). They have several years of experience in developing smart biomaterials essential for successful 4D bioprinting (task1.1).
- DemCon: Leading company in high-tech systems and materials. Their know-how is beneficial for *i)* implementation of conventional sensors in ELMs (WP1), *ii)* real-time monitoring in culture (WP2-4), *iii)* bringing products to market (WP5).
- Scinus Cell Expansion BV: Experts in real-time monitoring- and feedback-based control of *in vitro* culture systems for use in patients, knowledge of the regulatory environment (WP5).
- Long-term connections with patient's organisations, Dutch Liver patient Society, Dutch Arthritis Society and Princes Beatrix Muscle Foundation, that expressed their interest.

4.d Governance, organization and collaboration strategy of the consortium

ALIVE will be coordinated by Prof. Zadpoor with support of the two other leads. A dedicated project manager will be appointed (10%) to support organising and planning consortium meetings, composing progress reports, and managing a website. A consortium agreement will be drawn up before the start.

The leads and project manager (executive committee) are primarily responsible for monitoring whether tasks (see 3c) run on time, arrange with the management board adaptations if necessary to achieve the outputs (see 3b). The executive committee will stimulate the interaction between participants from different centers and harmonisation between the sub-projects.



Multidisciplinary teams will collaborate in each WP that meet monthly. To merge the disciplines, all hired personnel will have supervisors from different institutes. We will ensure cross fertilization between WPs by monthly meetings of WP-leaders and Programme leads that jointly form the ALIVE's management board. The entire programme (all participants) will meet twice a year. A kick-off meeting will be organised within 3 months after the start. The programme meetings are planned as live one-day meetings rotating between the 3 institutes. During these meetings all work will be presented in an interactive format. Educational lectures by PIs and demonstrations or lab tours will be provided to bring everyone to a certain level of understanding of the other disciplines. During these meetings, the progress, the future pathway as well as new grant options will be specifically discussed. Furthermore, an advisory committee led by Prof. Sarah Cartmell (Tissue Engineer; Manchester) with experts from science, clinics and patients will be composed. Committee members have already agreed to join. The advisory committee will attend the annual ALIVE meetings to provide feedback.

Annexes

I Budget table (including the external funding table)

Budget formulier: Convergentie Flagship Call

Flagship:					
Flagship acronym:	ALIVE				
Naam Penvoerder:					
Startdatum en looptijd van het pr	01/09/22	60.00	maanden	31/08/27	
Totaal project budget:	€ 1,999,408.00	plus matching	€	5,291,800.00	

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1\. Loonkosten:

Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedra	Kosten	year 1 (01/09/2022) in cash	year 1 in kind	year 2 (01/09/2023) in cash	year 2 in kind	year 3 (01/09/2024) in cash	year 3 in kind	year 4 (01/09/2025) in cash	year 4 in kind	year 5 (01/09/2026) in cash	year 5 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																				
TU Delft/Erasmus MC	postdoc	1.0	01/09/22	31/08/25	Schaal 10	€ 4,000	€ 216,000	€ 72,000		€ 72,000		€ 72,000						€ 216,000	€ 0	€ 216,000.00
TU Delft/Erasmus MC	postdoc	0.5	01/09/22	31/08/26	Schaal 10	€ 4,000	€ 144,000	€ 36,000		€ 36,000		€ 36,000		€ 36,000				€ 144,000	€ 0	€ 144,000.00
Erasmus MC/TUDelft	postdoc	1.0	01/01/23	31/12/24	Schaal 10	€ 4,000	€ 144,000	€ 48,000		€ 72,000		€ 24,000						€ 144,000	€ 0	€ 144,000.00
TU Delft/Erasmus MC	postdoc	1.0	01/06/24	31/05/26	Schaal 10	€ 4,000	€ 144,000			€ 24,000		€ 72,000		€ 48,000				€ 144,000	€ 0	€ 144,000.00
Erasmus MC/TUDelft	PhD	1.0	01/06/23	31/05/27	Schaal 7	€ 3,422	€ 246,384	€ 15,399		€ 61,596		€ 61,596		€ 61,596		€ 46,197		€ 246,384	€ 0	€ 246,384.00
Erasmus MC/TUDelft	PhD	1.0	01/06/23	31/05/27	Schaal 7	€ 3,422	€ 246,384	€ 15,399		€ 61,596		€ 61,596		€ 61,596		€ 46,197		€ 246,384	€ 0	€ 246,384.00
Erasmus MC/TUDelft	PhD	1.0	01/06/23	31/05/27	Schaal 7	€ 3,422	€ 246,384	€ 15,399		€ 61,596		€ 61,596		€ 61,596		€ 46,197		€ 246,384	€ 0	€ 246,384.00
Erasmus University / Erasmus MC	postdoc	1.0	01/01/24	30/06/25	Schaal 10	€ 4,000	€ 108,000			€ 48,000		€ 60,000						€ 108,000	€ 0	€ 108,000.00
Erasmus University/ TU Delft	postdoc	1.0	01/03/26	31/08/27	Schaal 10	€ 4,000	€ 108,000							€ 36,000		€ 72,000		€ 108,000	€ 0	€ 108,000.00
Erasmus MC	project manager	0.1	01/09/22	31/08/27	Schaal 11	€ 5,584	€ 50,256	€ 10,051		€ 10,051		€ 10,051		€ 10,051		€ 10,051		€ 50,256	€ 0	€ 50,256.00
						Totaal	€ 1,653,408		€ 212,248		€ 446,839		€ 458,839		€ 314,839		€ 220,642		Totaal loonkosten	€ 1,653,408

2\. Kosten van materialen en hulpmiddelen:

Partner die de kosten maakt	Materiaal / hulpmiddel		Kosten	year 1 in cash	year 1 in kind	year 2 in cash	year 2 in kind	year 3 in cash	year 3 in kind	year 4 in cash	year 4 in kind	year 5 in cash	year 5 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																
TU Delft	Bioprinter		€ 100,000	€ 100,000										€ 100,000	€ 0	€ 100,000.00
TU Delft	Materials		€ 56,000	€ 24,000		€ 14,000		€ 9,000		€ 9,000				€ 56,000	€ 0	€ 56,000.00
Erasmus MC	Materials		€ 148,000	€ 42,000		€ 32,000		€ 20,000		€ 38,000		€ 16,000		€ 148,000	€ 0	€ 148,000.00
Erasmus University	Materials		€ 5,000	€ -		€ 1,000		€ 1,000		€ 1,000		€ 2,000		€ 5,000	€ 0	€ 5,000.00
		Totaal	€ 309,000		€ 166,000		€ 47,000		€ 30,000		€ 48,000		€ 18,000		kosten materialen en hulpmiddelen	€ 309,000

3\. Kosten van gebruik van machines en apparatuur

Partner die de kosten maakt	Machine / apparatuur		Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	year 1 in cash	year 1 in kind	year 2 in cash	year 2 in kind	year 3 in cash	year 3 in kind	year 4 in cash	year 4 in kind	year 5 in cash	year 5 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																		
				Totaal	€ 0	#VERW!	#VERW!	#VERW!	#VERW!	#VERW!	#VERW!	#VERW!	#VERW!	#VERW!	#VERW!		taal kosten machines en apparatuur	€ 0

4\. Aan derden verschuldigde kosten

Partner die de kosten maakt	Naam derde, omschrijving kosten		Kosten	year 1 in cash	year 1 in kind	year 2 in cash	year 2 in kind	year 3 in cash	year 3 in kind	year 4 in cash	year 4 in kind	year 5 in cash	year 5 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																
		Totaal	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0		Totaal kosten derden	€ 0

5\. Publicatie, reis- en verblijfkosten

Partner die de kosten maakt	Naam derde, omschrijving kosten		Kosten	year 1 in cash	year 1 in kind	year 2 in cash	year 2 in kind	year 3 in cash	year 3 in kind	year 4 in cash	year 4 in kind	year 5 in cash	year 5 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																
TU Delft	conference visits postdocs		€ 5,000	€ 1,000.00		€ 1,500		€ 1,500.00		€ 1,000				€ 5,000	€ 0	€ 5,000.00
Erasmus MC	conference visits PhD students		€ 10,000			€ 2,500		€ 2,500.00		€ 2,500		€ 2,500.00		€ 10,000	€ 0	€ 10,000.00
Erasmus University	conference visits postdocs		€ 2,000					€ 1,000.00				€ 1,000.00		€ 2,000	€ 0	€ 2,000.00
TU Delft/Erasmus MC/ Erasmus University	organisation programme meetings, workshops and dissemination		€ 20,000	€ 4,000.00		€ 2,000		€ 2,000.00		€ 2,000		€ 10,000.00		€ 20,000	€ 0	€ 20,000.00
		Totaal	€ 37,000	€ 5,000		€ 6,000		€ 7,000		€ 5,500		€ 13,500		€ 37,000.00	al kosten derden	€ 37,000

6\. Totale kosten

	Kosten	Totaal per partner
Totale kosten	€ 1,999,408	€ 1,999,408

FINANCIERING - Bijdragen per partner per jaar

	in Euro	year 1 in cash	year 1 in kind	year 2 in cash	year 2 in kind	year 3 in cash	year 3 in kind	year 4 in cash	year 4 in kind	year 5 in cash	year 5 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties														
1	Erasmus MC	€ 644,600	€ 52,500	€ 661,300	€ 52,500	€ 200,000	€ 42,500	€ 200,000		€ 200,000		€ 1,905,900	€ 147,500	€ 2,053,400
2	TU Delft	€ 767,400		€ 841,000		€ 400,000		€ 500,000		€ 500,000		€ 3,008,400	€ 0	€ 3,008,400
3	Erasmus Universit	€ 50,000		€ 50,000								€ 100,000	€ 0	€ 100,000
	Totale bijdrage on	€ 1,462,000	€ 52,500	€ 1,552,300	€ 52,500	€ 600,000	€ 42,500	€ 700,000	€ 0	€ 700,000	€ 0	€ 5,014,300	€ 147,500	€ 5,161,800
Ondernemingen														
4	Cellink		€ 10,000		€ 10,000		€ 10,000		€ 10,000		€ 10,000		€ 50,000	€ 50,000
5	Poietis		€ 3,000		€ 3,000		€ 3,000		€ 3,000		€ 3,000		€ 15,000	€ 15,000
6	Regenhu		€ 5,000		€ 5,000		€ 5,000		€ 5,000		€ 5,000		€ 25,000	€ 25,000
	Demcon		€ 5,000		€ 5,000		€ 5,000		€ 5,000		€ 5,000		€ 25,000	€ 25,000
…	Scinus BV		€ 3,000		€ 3,000		€ 3,000		€ 3,000		€ 3,000		€ 15,000	€ 15,000
	Totale bijdrage on	€ 0	€ 26,000	€ 0	€ 26,000	€ 0	€ 26,000	€ 0	€ 26,000	€ 0	€ 26,000	€ 0	€ 130,000	€ 130,000
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			Totaal
	Totaal budget pe	€ 1,462,000	€ 78,500	€ 1,552,300	€ 78,500	€ 600,000	€ 68,500	€ 700,000	€ 26,000	€ 700,000	€ 26,000			€ 5,291,800
														Funding gap
														3,292,392
	Totaal budget	€ 1,540,500		€ 1,630,800		€ 668,500		€ 726,000		€ 726,000		€ 5,291,800		

Table explaining the matching by applicants

	matching for	Funding organisation	Title of the grant	PI	Year 1 PhD or PD	Year 1 in kind	Year 1 in cash*	Year 1 materials in cash	Year 2 PhD or PD	Year 2 in kind	Year 2 in cash*	Year 2 materials in cash
EUR	Task 5.3	Innovatiefonds Hagelunie	Horticulture Innovation	Van den Ende			€ 50,000				€ 50,000	
EMC	Task 1.4	NCOH	Bacteriophage Research on Immune response, Clinical use and Knowledge on virome changes	Wamel van, W.J.B.	1 PhD		€ 35,200	€ 8,000	1 PhD		€ 35,200	€ 8,000
	Task 1.4	Innovative Medicines Initiative (IMI) programme: New Drugs for Bad Bugs	Combatting Bacterial Resistance in Europa	Wamel van, W.J.B.	1 PD		€ 60,000	€ 40,000	1 PD		€ -	€ 40,000
	Task 5.1	ZonMw	BRAINMODEL	Bunnik, E.M.	1 PhD		€ -		1 PhD		€ -	
	Task 5.1	EU ERC AdV	VANGUARD	Bunnik, E.M.	1 PhD		€ -		1 PhD		€ -	
	Task3.1&3.3	CSC (in kind)	osteochondral repair	Van Osch	1 PhD	€ 15,000			1 PhD	€ 15,000		
	Task3.1	Chinese fellowship (in kind)	endochondral ossification	Farrell/ van Osch	1PhD	€ 15,000			1 PhD	€ 15,000		
	Task 3.2	Health-Holland TKI	HypOA	Van Osch/Farrell	1 PD		€ -	€ 10,000	1 PD		€ -	€ 5,000
	Task 3.1	Health-Holland TKI	BoneMend	Farrell	1 PhD		€ -		1 PhD		€ -	
	Task 3.1	Health~Holland	PhosphoNorm	van der Eerden, BCJ	1 PhD		€ -	€ 10,000	1 PhD		€ -	€ 5,000
	Task 4.1&4.3	LSH/NWO	Intense	De Zeeuw	2 PD, 1 PhD		€ -		2 PD, 1 PhD		€ -	
	Task 4.2&4.3	JPND/Zonmw	Neuromuscular junction-on a -Chip	Pijnappel	1 PhD		€ -	€ 3,000	1 PhD		€ -	€ 8,000
	Task4.1&4.4	Health Holland through Prinses Beatrix Spierfonds	Myofibre contractility in hiPSC-derived 3D muscle	Pijnappel	0.5 PhD		€ -	€ 7,500	0.5 PhD		€ -	€ 7,500
	Task 4.2&4.4	Health Holland through Prinses Beatrix Spierfonds	human skeletal muscle-on-chip model containing... progenitors and electrical impedance sensing	Pijnappel	0.5 PhD		€ -	€ 2,500	0.5 PhD		€ -	€ 15,000
	Task2.1&2.3	CSC (in kind)	Programmed cell death in liver diseases	Van der Laan/Verstegen	1.5 PhD	€ 22,500			1.5 PhD	€ 22,500		
	Task2.3	HUB	Collection of liver biopsies for research	Verstegen / vd Laan				€ 16,000				€ 16,000
	Task2.1&2.2	TKI Health Holland	The Matrix Reloaded	Verstegen / vd Laan	0.25 PhD		€ -	€ 9,108	0.25 PhD		€ -	€ 26,673
TUD	Task1.1&1.2	Interreg 2Seas	SiteDrug	Zadpoor, A.	1 PhD, 1PD		€ 100,000	€ 2,000	1 PhD, PD		€ 100,000	€ 1,500
	Task1.1	RMS foundation	Role of 3D scaffold architecture	Fratila-Apachitei, E.L.	1PD		€ -	€ 4,190	1 PD		€ -	€ 6,286
	Task1.1&1.2	NWO VIDI	Metallic Clay	Zadpoor, A.	1 PhD, 1 PD		€ 100,000	€ 8,000	1 PhD , 1PD		€ 75,000	€ 18,000
	Task1.2&1.4	NWO	DARTBAC	Zadpoor, A.	2 PhD, 0.5 PD		€ 110,000	€ 11,771	2 PhD, 0.5 PD		€ 110,000	€ 35,313
	Task1.3	EU ERC AdV	LoopingDNA	Dekker, C.	2 PD, 2PhD		€ -	€ 21,500	2 PD, 2 PhD		€ -	€ 65,000
	Task1.3	NWO ENW Groot	Guardians of protein disorder	Dekker, C.	1 PD, 2 PhD		€ -	€ 10,000	1 PD, 2 PhD		€ -	€ 30,000
				TOTAL		€ 52,500	€ 1,298,533	€ 163,569		€ 52,500	€ 1,265,200	€ 287,272
		* salary PhD € 40,000/year										
		* salary PD € 60,000/year										
Matching from grant proposals after the first 2 years												
EMC	WP2 and 3	EIC pathfinder (5y)	€ 1,000,000									
TUD	WP1 all tasks	EIC pathfinder (5y)	€ 2,500,000									

II Letters of intent and letters of commitment

CELLINK
A BICO COMPANY

March 2, 2022

LETTER OF INTENT
for the
Adaptive Living Engineered organs (ALIVE) PROJECT

Dear Prof. Zadpoor,

I, Itedale Namro Redwan, in my capacity of CSO at CELLINK hereby confirm that CELLINK intends to contribute to the Adaptive Living Engineered organs (ALIVE) project, as applied for by the main applicant, Amir Zadpoor, Full Professor at Biomechanical Engineering Department, Delft University of Technology (TU Delft).

CELLINK is a world-leading company in 3D bioprinting and the innovator of bioinks and technologies at the forefront of making on-demand bioprinted tissues a reality. Our team of scientists, engineers and entrepreneurs ensure that processing routes are safe, robust, and optimized. Together with our collaborators in hundreds of labs in more than 50 countries, we work side by side to ensure quality and support. Our drive to create an impact aims to open the possibility for more extensive medical research and pave the way for future healthcare applications.

CELLINK strongly supports the 'Adaptive Living Engineered organs (ALIVE)' project. We will provide in-kind contributions over a period of five years, including:

Item or activity	Amount
Support for preselection of objective	4.000 €
Basic tests at Nanoscribe of chosen objective	10.000 €
Annual tele/online support for developing adapted printing recipe	7.500 €
Online user training on Medium feature (25x objectives) and Small feature (63x) solution set	2.000 €
Advanced personalized online user training on validating and creating print recipes with Small Feature (63x) Solution Set and new objective	3.000 €
General annual support for market transfer and development	10.000 €
Annual service support for system and process development	10.000 €
International webinars for outreach and impact of this project	3.500 €
Total	50.000 €

We will ensure that the results of the project will be disseminated is a mutually agreeable fashion respecting the intellectual property rights of all partners involved.

Yours sincerely

Itedale Namro Redwan

Itedale Namro Redwan, PhD
CSO CELLINK

Dr. Michael Thiel
CSO & Co-Founder Nanoscribe GmbH

Långfilsgatan 7
412 76 Gothenburg
Sweden

155 Seaport Blvd, Unit 2B
Boston, MA 02210, USA

EU Phone: +46 (0)31 128 700
US Phone: (+1) 833 235 5465

Email: sales@cellink.com
Web: www.cellink.com

Pessac, France

February 07, 2022

LETTER OF INTENT

for the ***Adaptive LIving Engineered organs (ALIVE) PROJECT***

Dear Dr. Mohammad J. Mirzaali, PhD, Assistant Professor
Delft University of Technology

I, Bruno Brisson, in my capacity of General Manager and VP Business Development at Poietis, hereby confirm that POoietis intends to contribute to the Adaptive LIving Engineered organs (ALIVE) project, as applied for by the main applicant, Dr. Amir Zadpoor, Full Professor at Biomechanical Engineering Department, Delft University of Technology (TUDelft).

In particular, we are interested in contributing to your project where printing technology is developed for the engineering of 3 dimensional tissue modules.

POoietis intends to support the project providing an in-kind contribution representing a monetary value of € 15 000 as it will be detailed in the project proposal and budget form.

More specifically, Poietis will invest 125 man hrs of a senior research employee (25 hours per year of the project) valued at € 120 per hour, who will closely work with the academic partners in providing technological support and advice to the consortium. In particular this contribution will facilitate the academic partners in the use of Poietis' high-end laser-based integrated bioprinter for the project goals.

Sincerely,

____

Name: Bruno Brisson

Position: Co-founder, VP Business Development

Date: February 07th, 2022

Siège social : S.A.S Poietis
Bioparc Bordeaux Métropole - Bâtiment C- 27 allée Charles Darwin - 33600 Pessac – France
SIRET: 804 776 797 00024
Contact : contact@poietis.com – Website : www.poietis.com

REGENHU
EVOLVE.

Dr Simon MacKenzie
Chief Executive Officer
ZI du Vivier 22
CH-1690 Villaz-St-Pierre

Villaz-St-Pierre, February 10th 2022

LETTER OF INTENT
for the
Adaptive LIving Engineered organs (ALIVE) ***PROJECT***

Dear Prof. Amir A. Zadpoor,

I, Simon MacKenzie, in my capacity of CEO at REGENHU LTD hereby confirm that REGENHU intends to contribute to the Adaptive LIving Engineered organs (ALIVE) project, as applied for by the main applicant, Amir Zadpoor, Full Professor at Biomechanical Engineering Department, Delft University of Technology (TUDelft). We confirm having expertise in providing equipment allowing the creation of various types of large, multicellular tissues with complex 3D architectures (i.e., liver, cartilage, bone, muscle and brain), essential to the success of the ALIVE project. For example, our equipment and the combination of its unique technologies in a single process have enabled the success of the H2020 funded ORGANTRANS Project with the aim to achieve the controlled assembly of live spheroid containing liver organoids into a bioconstruct (www.organtrans.eu)

REGENHU LTD intends to provide an in-kind contribution of 20 hours per year (100 hours within the total project for the following tasks:

- Configure and supply bioprinting instruments & software (R-GEN SERIES) to any of the participants according to their respective budgets available.
- Help the participants to select dispensing tools, process tools, workzones, options and accessories in order to best address the different application challenges
- Provide consultancy support from our application engineers, training team and scientific advisors, to increase the success of the project by providing training, assist in building and refining applications, using our 14 years of expertise in Bioprinting (advice in setting and optimizing parameter definitions, provide scientific support and introductions to other users of relevance from the REGENHU user network, extended training in the use of the instrument and software).

This represents a monetary value of 25000 € and can be further detailed in the project proposal and budget form.

Yours sincerely,

S.J. MacKenzie

Name:	Dr Simon MacKenzie
Position:	Chief Executive Officer (REGENHU)
Date:	10th February 2022

Letter of Intent for a project partner

To:
Prof. dr. A.A. Zadpoor
Department of Biomechanical Engineering,
Faculty of Mechanical, Maritime, and
Materials Engineering
Delft University of Technology (TU Delft)
Mekelweg 2, 2628CD, Delft
The Netherlands

SCINUS CELL EXPANSION

Subject: Letter of intent for the Adaptive LIving Engineered organs (ALIVE) ***PROJECT***

Bilthoven, 14 March 2022

Dear Prof. Zadpoor,

I, Michiel Jannink, in my capacity of CEO at Scinus Cell Expansion Netherlands BV hereby confirm that Scinus Cell Expansion Netherlands BV intends to contribute to the Adaptive LIving Engineered organs (ALIVE) project, as applied for by the main applicant, Amir Zadpoor, Full Professor at Biomechanical Engineering Department, Delft University of Technology (TUDelft).

Scinus Cell Expansion Netherlands BV intends to provide an in-kind contribution dedicated to activities in the field of sensor integration in bioreactors, real-time control/monitoring in cell culture, regulatory considerations and market perspectives, representing a monetary value of € 15.000 and further detailed in the project proposal and budget form.

Yours sincerely,

Michiel Jannink
CEO
Scinus Cell Expansion Netherlands BV

Support letters Patient organisations

Beschermvrouwe
Prinses Beatrix

ReumaNederland

Admiraal Helfrichstraat 1
1056 AA Amsterdam
Postbus 59091
1040 KB Amsterdam

t 020 • 589 64 64

info@reumanederland.nl
www.reumanederland.nl

KvK 40408531
IBAN NL64 ABNA 0433 533 633

Erasmus Medical Center
T.a.v. prof.dr. Gerjo van Osch
Doctor Molenwaterplein 40
3015 GD Rotterdam

Date:	February 24 2022	Contact:	Heidi van Vugt
Reference:	HvV22003	E-mail:	h.vanvugt@reumanederland.nl
Subject:	Commitment letter to "Adaptive LIving Engineered organs (ALIVE)"		

Dear prof.dr. van Osch,

With this letter I declare on behalf of ReumaNederland (Dutch Arthritis Society) our interest in and great support of the project idea described in "Adaptive LIving Engineered organs (ALIVE)" project, as applied for by the main applicant, Amir Zadpoor, Delft University of Technology (TUDelft).

The project idea fits very well with the mission of ReumaNederland that in 2040 it's possible to cure osteoarthritis. The project will focus on the development of new technologies to help patients with cartilage and bone defects as a result of osteoarthritis. There are over 1.5 million people in the Netherlands with osteoarthritis and we currently have no treatment to cure them. The program ALIVE aligns also with our recently defined alliance "Fight Osteoarthritis" and in particular the combination of technical, biological, medical but also the involvement of groups from humanities, business and social sciences is a strength of this program. We believe that it is important that different organisations and experts working together on such a problem as osteorthritis and that shall lead faster to a solution than working on their own.

ReumaNederland is a patient organisation that represents over 2.000.000 people suffering from rheumatic diseases, such as rheumatoid arthritis and osteoarthritis. ReumaNederland helps patients to improve their quality of live -in a wider perspective- with rheumatic diseases. Together with patients, ReumaNederland advocates patients' rights in politics and care. ReumaNederland works together with local patient associations and with nationwide more disease specific patient associations (amongst them is a patient association for osteoarthritis). Next to patient advocacy, ReumaNederland aims to improve patient care in the Netherlands. Scientific research is another main theme, either by financing research or by setting the agenda for specific topics/themes. The ultimate goal is cure and better care. We feel that the project of Amir Zadpoor shall contribute to get more insight into the development of a treatment for osteoarthritis.

Yours sincerely,

Corne Baatenburg de Jong
Vice president, ReumaNederland

Nederlandse Leverpatiënten Vereniging

Van Boetzelaerlaan 24 J
3828 NS Hoogland
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KvK Gooi-, Eem en Flevoland 40535249
ANBI-nummer: 808313472

TU Delft
t.a.v. prof. dr. A. Zadpoor
Voorzitter ALIVE consortium
Biomechanical Engineering
Postbus 5
2600 AA Delft

Hoogland 22 februari 2022
Ref: 2022.02.35/JW

Geachte professor Zadpoor,

Met grote belangstelling heeft de Nederlandse Leverpatienten Vereniging (NLV) kennis genomen van het voornemen van het Erasmus MC een nauw samenwerkingsverband aan te gaan met de Technische Universiteit Delft en de Erasmus Universiteit, gericht op *Health and Technology*. De plannen voor een convergence wetenschapsgemeenschap (Flagship), samen met genoemde instituten verheugen ons zeer. Wij zijn ervan zijn overtuigd dat dit alle processen rondom de zorg voor en van de leverpatiënt zal bespoedigen.

De NLV is al bij vele ontwikkelingen betrokken die betrekking hebben op diagnostiek, behandelinterventies, evenals de ontwikkelingen bij levertransplantatie waar veel leverpatiënten uiteindelijk mee in aanraking zullen komen. Wij weten dat er al goede ontwikkelingen gaande zijn op het gebied van tissue engineering ten einde leverweefsel in het laboratorium te maken en in te zetten voor ziekte modellering of op de langere termijn voor transplantatiedoeleinden. Een samenwerkingsverband zullen onder meer deze en nieuwe ontwikkelingen alleen maar ten goede komen.

Alleen door samenwerking met alle facetten van de wetenschap kan het uiteindelijke doel worden bereikt: het verbeteren van de kwaliteit van leven voor patiënten. Kortom, wij steunen dit fantastische initiatief van harte.

Met vriendelijke groet,
mede namens het bestuur

JWillemse

drs. José A. Willemse
directeur

III Letter of support from department head of the main lead

Date 9-03-2022
Contact person H. E. J. Veeger
Telephone/fax +31-15-2783133
E-mail H.E.J.Veeger@tudelft.nl
Subject Letter of support for Adaptive LIving Engineered organs (ALIVE) PROJECT

TU Delft

Delft University of Technology

Faculty of Mechanical,
Maritime and Materials Engineering
Department of Biomechanical Engineering

Address:
Mekelweg 2
2628 CD Delft
The Netherlands

Dear Prof. Zadpoor,

In my capacity as the Chair of the Biomechanical Engineering Department, Delft University of Technology (BME-TUDelft), I hereby confirm that BME-TUDelft intends to contribute to the Adaptive LIving Engineered organs (ALIVE) project, as applied for by the main applicant, Prof. Amir A. Zadpoor. The topic of this proposal is one of the important and strategic areas for our department and our partners at Erasmus MC and Erasmus University in the coming years for which we have already made significant investments in terms of state-of-the-art facilities. We aim to further invest in this exciting and novel area in the coming years and hope that the convergence program between TUDelft and Erasmus MC can support our ambitions for establishing an exciting and unique research direction.

Yours sincerely

H. E. J. Veeger
Professor
Department Chair, BME-TUDelft

Page / of 1/1

IV References

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